Lab Sheet 4: SQL Operators

Arithmetic Operators

1. Perform addition of two numbers using +.

```
116 •  SELECT 20 + 10 AS Addition;
```

Result Grid | Filter Rows:

Addition
30

2. Perform subtraction using -.

```
117 •  SELECT 20 - 10 AS Subtraction;
```

Result Grid | Filter Rows:

Subtraction
10

3. Perform multiplication using *.

```
118 •  SELECT 20 * 10 AS Multiplication;
```

Result Grid | Filter Rows:

Multiplication
200

4. Perform division using /.

```
119 •  SELECT 20 / 10 AS Division;
```

Result Grid | Filter Rows:

Division
2.0000

5. Calculate the total amount as Price × Quantity.

```
120 •  SELECT Product_Name, Price, Quantity,Price * Quantity AS Total_Amount FROM Product
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product_Name	Price	Quantity	Total_Amount
Laptop	57750.00	10	577500.00
Mouse	840.00	25	21000.00
Keyboard	1575.00	15	23625.00
Monitor	12600.00	0	0.00
Printer	15750.00	5	78750.00

Logical Operators

6. Demonstrate AND with two conditions.

```
121 •  SELECT * FROM Student WHERE Gender = 'Female' AND Age >= 20;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

7. Demonstrate OR with two conditions.

```
122 •  SELECT * FROM Student WHERE Address = 'Jaipur' OR Address = 'Delhi';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

8. Demonstrate NOT with a condition.

```
123 •  SELECT * FROM Student WHERE NOT Gender = 'Male';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

9. Find students belonging to a specific course **AND** city.

```
124 •  SELECT * FROM Student WHERE Course = 'B.Tech CSE' AND Address = 'Jaipur';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap C

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

10. Find employees belonging to either of two departments.

```
SELECT * FROM Employe WHERE Department = 'IT' OR Department = 'HR';
```

Employee_ID	Name	Salary	Department	City
11	Amit	35000.00	IT	Delhi
12	Rahul	42000.00	HR	Jaipur
14	Priya	38000.00	IT	Pune
15	Rohit	28000.00	HR	Kota

Comparison Operators

11. Demonstrate the = operator.

```
SELECT * FROM Student WHERE Course = 'BCA';
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
141	Arjun	NULL	21	NULL	BCA	NULL	NULL

12. Demonstrate the > operator.

```
SELECT * FROM Student WHERE Percentage > 80;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

13. Demonstrate the < operator.

```
SELECT * FROM Student WHERE Percentage < 80;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215

14. Demonstrate >= and <=.

```
SELECT * FROM Student WHERE Percentage >= 80 AND Percentage <= 90;
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216

15. Demonstrate the <>/!= operator.

```
SELECT * FROM Student WHERE Course != 'BCA';
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

Special Operators

16. Demonstrate BETWEEN for salary range

```
SELECT * FROM Employe WHERE Salary BETWEEN 30000 AND 45000;
```

Employee_ID	Name	Salary	Department	City
11	Amit	35000.00	IT	Delhi
12	Rahul	42000.00	HR	Jaipur
14	Priya	38000.00	IT	Pune

17. Demonstrate IN for selecting multiple cities

```
SELECT * FROM Student WHERE Address IN ('Jaipur', 'Delhi', 'Mumbai');
```

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
103	Jayshree	Female	21	Delhi	B.Tech IT	85.50	9876543276
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250

18. Demonstrate NOT IN.

```
132 • SELECT * FROM Student WHERE Address NOT IN ('Jaipur', 'Delhi');
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
102	Radha	Female	10	Kashipur	B.Tech IT	85.50	9876543276
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216

19. Demonstrate LIKE for names beginning with a particular letter.

```
133 • SELECT * FROM Student WHERE Name LIKE 'A%';
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
101	Anshika	Female	20	Delhi	B.Tech CSE	85.50	9876543210
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213
114	Aditi	Female	21	Jaipur	B.Tech CSE	90.50	9876543250
141	Arjun	NULL	21	NULL	BCA	NULL	NULL

20. Demonstrate IS NULL and IS NOT NULL.

```
134 • SELECT * FROM Student WHERE Percentage IS NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Imp

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo
141	Arjun	NULL	21	NULL	BCA	NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Set Operations

21. Demonstrate UNION using two compatible queries.

```
135 • SELECT Name FROM Student WHERE Address = 'Jaipur'
136 UNION
137 SELECT Name FROM Student WHERE Address = 'Delhi';
```

Result Grid | Filter Rows: | Export: | Wrap Cel

Name
Shreya
Aditi
Anshika
Jayshree
Rahul

22. Demonstrate UNION ALL.

```
138 • SELECT Name FROM Student WHERE Address = 'Jaipur'
139 UNION ALL
140 SELECT Name FROM Student WHERE Address = 'Delhi';
```

Result Grid | Filter Rows: | Export: | Wrap Cel

Name
Shreya
Aditi
Anshika
Jayshree
Rahul

23. Demonstrate INTERSECT where supported by the DBMS

```
141 • SELECT Name FROM Student
142 WHERE Percentage >= 80 AND Age >= 20;
```

Result Grid | Filter Rows: | Expor

Name
Anshika
Jayshree
Ruhani
Shreya
Maya
Priya
Neha
Simran
Aditi

24. Demonstrate EXCEPT/MINUS where supported.

```
143 • SELECT Name FROM Student WHERE Address = 'Jaipur'
144 AND Course <> 'BCA';
```

Result Grid | Filter Rows: | Export: | Wrap Cel

Name
Shreya
Aditi

25. Perform a suitable problem using multiple operators together, such as **Arithmetic + Logical + Comparison + Special operators**.

```
145 • SELECT Product_Name, Price, Quantity,
146 Price * Quantity AS Total_Amount FROM Product WHERE Price * Quantity > 10000
147 AND Quantity BETWEEN 5 AND 25;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product_Name	Price	Quantity	Total_Amount
Laptop	57750.00	10	577500.00
Mouse	840.00	25	21000.00
Keyboard	1575.00	15	23625.00
Printer	15750.00	5	78750.00